

MEGA-CAP TECHNOLOGY

FULL EQUITY RESEARCH REPORT

Mega-Cap Technology Drawdown and Accumulation Framework

A Business-Fundamentals and Historical-Risk Study of Long-Term Returns, Drawdown Frequency, Recovery Characteristics, and Staged Buying Strategies

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Independent research on long-term returns, drawdown behavior, recovery characteristics and staged accumulation.

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Financial data current through the latest company reports available as of August 12, 2026. Historical price analysis covers complete calendar years through 2025.

Executive Summary

This report develops a company-specific accumulation framework for nine mega-cap technology and technology-adjacent companies: NVIDIA, Tesla, Broadcom, Meta Platforms, Apple, Taiwan Semiconductor Manufacturing Company, Amazon, Microsoft, and Alphabet.

The study combines two different types of evidence.

First, it examines the latest operating composition of each company: what the company sells, which businesses produce most of its revenue, which segments are growing fastest, and what variables now determine the durability of the investment thesis. This matters because historical price statistics cannot distinguish a temporary market correction from a permanent deterioration in a business.

Second, it analyzes adjusted daily share-price history over complete calendar years from 2010 through 2025. The historical study measures compound annual growth, positive-year consistency, annual maximum drawdown, total and downside volatility, typical one- and three-month losses, recovery time, and the frequency of 20%, 30%, 40%, and 50% annual drawdowns.

Together, these two layers answer four practical questions:

1. What economic engine is the investor buying?
2. How attractive has the stock's long-term return profile been?
3. How unusual is the current drawdown relative to the stock's own history?
4. Is the business thesis sufficiently intact to justify increasing exposure?

The latest financial reports show that these companies are less economically similar than their “mega-cap technology” label suggests. Meta still receives 97.6% of revenue from advertising. NVIDIA receives 92.1% from Data Center. Apple receives almost half of revenue from the iPhone, while Services contributes 28.1%. Alphabet still receives 52.8% from Search, but Google Cloud has reached 20.7% of revenue and is growing much faster. Amazon's retail segments generate most of its sales, while AWS generates approximately 60% of operating income. Broadcom combines a rapidly growing AI semiconductor franchise with a slower-growing infrastructure-software business. TSMC is now led by high-performance computing rather than smartphones. Tesla remains primarily an automotive company, although Energy and Services are becoming more important. Microsoft has the most balanced enterprise platform, with Intelligent Cloud and Productivity and Business Processes jointly accounting for most revenue.

The historical return analysis also shows very different investment paths. NVIDIA, Tesla, and Broadcom generated the highest historical CAGRs, but NVIDIA and Tesla required far greater tolerance for volatility. Microsoft produced the lowest CAGR in the high-quality group, but it also produced the shallowest typical annual drawdown, the lowest downside volatility, and one of the shortest median recoveries. Broadcom delivered the strongest historical return relative to its median annual drawdown. Tesla delivered exceptional long-term appreciation but the most difficult holding experience: the deepest median annual drawdown, the highest total and downside volatility, the worst short-term losses, and the longest median recovery.

The drawdown-frequency analysis is the bridge between statistics and action. A 20% Microsoft drawdown occurred in only 25% of observed calendar years. A 20% NVIDIA or Broadcom drawdown occurred in 68.8% of years. Every complete Tesla year in the sample contained at least a 20% drawdown. Therefore, a universal “buy every stock after it falls 20%” rule would systematically deploy capital too early in the highest-beta names.

The resulting portfolio framework is:

- **Core long-duration compounders — Microsoft, Alphabet, and Apple.** Maintain continuous exposure. A 20% drawdown is a meaningful acceleration point; approximately 30% is a major accumulation zone if the thesis remains intact.
- **Opportunistic compounders — Broadcom, Meta, and Amazon.** Maintain baseline positions but retain more dry powder. Approximately 30% is the first major accumulation zone; 40% requires a stronger fundamental review but can offer exceptional expected returns if the decline is cyclical or sentiment-driven.
- **Cyclical compounder — TSMC.** Maintain long-term exposure, but combine price thresholds with semiconductor

utilization, leading-edge demand, capital expenditure, and geopolitical analysis. Approximately 30% is the most important historical accumulation threshold.

- **High-beta compounder — NVIDIA.** A 20% decline is normal. Approximately 30% is the first meaningful add zone and 40% is the major accumulation zone, subject to AI capital-spending and competitive validation.
- **Optionality asset — Tesla.** A 20%–30% decline is ordinary historical behavior. Major incremental capital should generally be reserved for 40%–50% drawdowns after reassessing automotive economics and the value of FSD, Robotaxi, Optimus, and Energy.

The central conclusion is not that historical drawdowns mechanically predict future bottoms. It is that investment intensity should rise only when three conditions align: the drawdown is unusually deep for that specific stock, the business thesis remains intact, and valuation offers a materially improved forward return.

1. Research Design and Methodology

1.1 Company Sample and Price Data

Company	Ticker	Core historical sample
NVIDIA	NVDA	2010–2025
Tesla	TSLA	2011–2025
Broadcom	AVGO	2010–2025
Meta Platforms	META	2012–2025
Apple	AAPL	2010–2025
Taiwan Semiconductor Manufacturing	TSM	2010–2025
Amazon	AMZN	2010–2025
Microsoft	MSFT	2010–2025
Alphabet	GOOGL	2010–2025

Adjusted daily prices are used so that stock splits and cash distributions are reflected in the return series. The main comparison uses only complete calendar years. Tesla's incomplete 2010 IPO year and the incomplete 2026 calendar year are excluded. Meta is measured from its public-market history beginning in 2012.

The business analysis uses each company's latest available quarterly report as of August 12, 2026. Fiscal calendars differ, so the latest quarters are not perfectly synchronized. Revenue share always refers to the company's own latest reported quarter.

1.2 Why Business Fundamentals Must Precede Drawdown Analysis

Price history describes how a stock behaved; it does not explain why it fell or whether it deserves to recover. A 40% correction caused by higher market discount rates is economically different from a 40% decline caused by lost market share, collapsing unit economics, balance-sheet stress, or technological displacement.

The framework therefore applies drawdown statistics only after evaluating:

- the company's present revenue engine;
- growth and profitability of the key segment;
- customer and product concentration;
- capital intensity and free-cash-flow conversion;
- competitive or regulatory threats; and
- valuation after the decline.

1.3 Core Historical Measures

CAGR is the annualized rate at which the adjusted share price compounded between the beginning and end of the sample:

$$\text{CAGR} = (\text{Ending Price} / \text{Starting Price})^{(1 / \text{Years})} - 1$$

Positive Years is the percentage of complete calendar years with a positive adjusted-price return. It measures consistency, not intra-year risk.

Median Annual Maximum Drawdown is the median of each calendar year's deepest peak-to-trough decline. It represents the stock's typical worst drawdown within a year.

Annualized Volatility annualizes the standard deviation of daily returns using approximately 252 trading days. It treats upside and downside moves symmetrically.

Downside Volatility focuses on negative-return observations. It is generally more relevant to position sizing because it better captures the severity of falling markets.

Typical Worst 1M and 3M are the median, across calendar years, of each year's worst rolling one-month and three-month return. They approximate the short-term loss an investor could experience after buying near a local high.

Median Recovery is the median number of trading days from a prior peak until the adjusted price regains that peak. It measures the duration of the complete underwater period, not merely the time from trough to recovery.

Historical Drawdown Recurrence converts the number of qualifying calendar years into an intuitive interval:

Recurrence Interval = Complete Observation Years / Years Meeting the Threshold

Annual Historical Probability expresses the same count as a percentage:

Historical Frequency = Years Meeting the Threshold / Complete Observation Years

These are descriptive sample statistics, not forecasts or guarantees.

2. Current Business Mix and Growth Analysis

2.1 NVIDIA — AI Infrastructure Concentration

NVIDIA's economic center has shifted decisively from gaming graphics to AI infrastructure. It sells accelerated-computing GPUs, complete AI systems, high-speed networking, CUDA software, enterprise AI tools, gaming and workstation processors, and computing platforms for vehicles and robotics.

Latest report: FY2027 Q1, quarter ended April 26, 2026

Business	Revenue	Revenue share	YoY growth
Data Center	\$75.2B	92.1%	92%
Edge Computing	\$6.4B	7.8%	29%
Total	\$81.6B	100%	85%

Under the previous disclosure, Data Center Compute reached \$60.4B, up 77%, while Data Center Networking reached \$14.8B, up 199%. NVIDIA now reports Gaming, workstation, automotive, robotics, and other edge applications together as Edge Computing.

The investment implication is concentration. NVIDIA has extraordinary exposure to global AI infrastructure spending. This creates very high operating leverage while demand is expanding, but also increases sensitivity to hyperscaler CapEx, GPU utilization, HBM and advanced-packaging availability, custom ASIC competition, and the return customers ultimately earn on AI infrastructure.

2.2 Tesla — Automotive Core with Emerging Energy and Services

Tesla sells electric vehicles and leases, regulatory credits, charging and vehicle software, Megapack and Powerwall storage products, solar products, insurance, used vehicles, maintenance, and charging services. FSD, Robotaxi, Optimus, and AI infrastructure contribute heavily to valuation expectations but remain less important to reported revenue than the automotive business.

Latest report: 2026 Q2, quarter ended June 30, 2026

Business	Revenue	Revenue share	YoY growth
Automotive	approximately \$20.52B	approximately 72.7%	approximately 23%
Energy Generation and Storage	\$3.14B	11.1%	13%
Services and Other	\$4.58B	16.2%	50%
Total	\$28.24B	100%	approximately 26%

Tesla remains primarily an automobile manufacturer by current revenue. Energy and Services are growing in importance, but a large part of the equity value still depends on the future economics of autonomy and robotics. Tesla's thesis therefore requires separating demonstrated revenue from long-duration optionality.

2.3 Broadcom – AI Semiconductors Plus Infrastructure Software

Broadcom supplies custom AI accelerators, data-center networking chips, switching and connectivity products, broadband, wireless and storage semiconductors, together with VMware virtualization, private-cloud, security, and mainframe software.

Latest report: FY2026 Q2, quarter ended May 3, 2026

Business	Revenue	Revenue share	YoY growth
Semiconductor Solutions	\$15.01B	67.6%	79%
Infrastructure Software	\$7.18B	32.4%	9%
Total	\$22.19B	100%	48%

AI semiconductor revenue was \$10.8B, up 143%, representing approximately 48.7% of total company revenue. Broadcom combines a rapidly growing but customer-concentrated AI semiconductor franchise with a slower-growing, recurring software cash-flow base. The most important risks are hyperscaler concentration, the durability of custom-accelerator programs, AI networking share, VMware customer retention, and leverage.

2.4 Meta Platforms – Advertising Economics Fund the AI and Reality Labs Bets

Meta operates Facebook, Instagram, Messenger, WhatsApp, and Threads. Its economic engine remains performance advertising. Reality Labs includes virtual- and augmented-reality hardware, software, and content.

Latest report: 2026 Q2, quarter ended June 30, 2026

Business	Revenue	Revenue share	YoY growth
Advertising	\$59.36B	97.6%	27.5%
Family of Apps other revenue	\$1.01B	1.7%	72.7%
Reality Labs	\$0.43B	0.7%	16.5%
Total	\$60.80B	100%	28.0%

Family of Apps produced \$60.37B of revenue, or 99.3% of the company total. Reality Labs generated an operating loss of \$4.62B. Meta's AI investment currently matters most because recommendation and advertising models can improve engagement, impressions, pricing, and advertiser return. The key question is whether AI CapEx strengthens the advertising engine quickly enough to offset rising depreciation and experimental-business losses.

2.5 Apple – iPhone Scale with a High-Margin Services Layer

Apple sells the iPhone, Mac, iPad, wearables and accessories, and a growing set of services including the App Store, iCloud, Apple Music, Apple TV+, AppleCare, payments, and licensing.

Latest report: FY2026 Q3, quarter ended June 27, 2026

Business	Revenue	Revenue share	YoY growth
iPhone	\$54.25B	49.6%	21.7%
Services	\$30.74B	28.1%	12.1%

Business	Revenue	Revenue share	YoY growth
Mac	\$10.35B	9.5%	28.7%
Wearables, Home and Accessories	\$7.88B	7.2%	6.5%
iPad	\$6.19B	5.7%	-5.9%
Total	\$109.42B	100%	16.4%

Apple still receives approximately half of revenue from the iPhone, but Services now provides a large recurring, high-margin layer on top of the installed base. The major thesis variables are ecosystem retention, replacement cycles, Services monetization and regulation, China exposure, AI competitiveness, and valuation.

2.6 TSMC — HPC Replaces Smartphones as the Primary Platform

TSMC manufactures chips designed by customers such as NVIDIA, Apple, AMD, and Broadcom. Its competitive position rests on leading-edge process technology, manufacturing yield, scale, customer trust, and increasingly advanced packaging.

Latest report: 2026 Q2, quarter ended June 30, 2026

Platform	Revenue share	QoQ growth
High-Performance Computing	66%	20%
Smartphone	22%	-4%
Internet of Things	5%	4%
Automotive	4%	15%
Digital Consumer Electronics	1%	5%
Other	2%	5%

Total revenue increased 36% year over year and 12% sequentially. Advanced nodes at 7nm and below accounted for 77% of wafer revenue. TSMC is no longer primarily a smartphone-growth story: HPC has increased to 66% of revenue, while smartphones have declined to 22%. The main risks are advanced-node execution, AI demand, customer inventory, overseas-fab economics, capital intensity, concentration, and Taiwan geopolitical risk.

2.7 Amazon — Retail Produces Scale; AWS Produces Disproportionate Profit

Amazon operates North American and international e-commerce, third-party seller services, logistics, Prime subscriptions, advertising, physical stores, and AWS cloud computing.

Latest report: 2026 Q2, quarter ended June 30, 2026

Segment	Revenue	Revenue share	YoY growth
North America	\$116.2B	57.9%	16%
International	\$42.2B	21.0%	15%
AWS	\$42.2B	21.0%	37%
Total	\$200.6B	100%	20%

AWS generated \$16.6B of the company's \$27.5B operating income, or approximately 60%. Advertising grew 26%, although Amazon includes advertising revenue inside its geographic segments rather than reporting it as a separate operating segment. The investment thesis rests on AWS growth and margins, retail efficiency, advertising monetization, international profitability, and whether AI infrastructure investment ultimately converts into free cash flow.

2.8 Microsoft — The Most Balanced Enterprise Technology Platform

Microsoft combines Microsoft 365, Teams, LinkedIn, Dynamics, Azure, server products, GitHub, Windows, devices, Xbox, and search advertising. Its portfolio is diversified across enterprise productivity, cloud infrastructure, developer tools, security, consumer software, and gaming.

Latest report: FY2026 Q4, quarter ended June 30, 2026

Segment	Revenue	Revenue share	YoY growth
Intelligent Cloud	\$39.3B	43.7%	32%
Productivity and Business Processes	\$37.8B	42.0%	14%
More Personal Computing	\$12.9B	14.3%	-4%
Total	approximately \$90.0B	100%	18%

Azure and other cloud services grew 43%. Microsoft Cloud revenue reached \$59.3B, up 27%. Microsoft 365 Commercial cloud grew 14% on a reported basis, LinkedIn grew 12%, and Dynamics 365 grew 13%. Windows OEM and Devices fell 7%, while Xbox content and services fell 10%.

Microsoft's diversification and recurring enterprise revenue help explain its historical price stability. The current tension is between accelerating Azure and Copilot demand and the enormous capital expenditure required to supply AI capacity. The thesis increasingly depends on utilization, pricing, margins, and free-cash-flow returns on AI infrastructure.

2.9 Alphabet — Search Funds a Rapidly Scaling Cloud Business

Alphabet operates Google Search, YouTube, advertising networks, subscriptions and devices, Google Cloud, and Other Bets such as Waymo.

Latest report: 2026 Q2, quarter ended June 30, 2026

Business	Revenue	Revenue share	YoY growth
Google Search and Other	\$63.27B	52.8%	16.8%
Google Cloud	\$24.77B	20.7%	81.8%
Subscriptions, Platforms and Devices	\$12.91B	10.8%	15.2%
YouTube Ads	\$11.06B	9.2%	12.9%
Google Network	\$7.30B	6.1%	-0.7%
Other Bets	\$0.38B	0.3%	2.4%
Total	\$119.80B	100%	24.2%

Search remains the largest revenue source, but Cloud has become the second engine and the fastest-growing major segment. Google Cloud operating income rose to \$8.81B. The central question is whether generative AI strengthens Search usage and monetization or gradually changes the economics of information discovery. Cloud growth, Cloud margins, YouTube, AI Search monetization, CapEx, and free cash flow are therefore the critical variables.

2.10 Cross-Company Business Comparison

Company	Dominant economic engine	Concentration	Fastest important growth driver	Principal fundamental risk
NVDA	AI Data Center	92.1%	Networking and AI compute	AI CapEx/utilization cycle
TSLA	Automotive	approximately 72.7%	Services; long-term autonomy optionality	Auto margins and optionality valuation
AVGO	Semiconductors	67.6%	AI custom silicon/networking	Customer concentration and AI cycle
META	Advertising	97.6%	AI-enhanced ads	CapEx and platform/regulatory risk
AAPL	iPhone	49.6%	Mac and iPhone in latest quarter	Replacement cycle and ecosystem regulation
TSM	HPC manufacturing	66%	AI/HPC and advanced nodes	Cycle and geopolitics
AMZN	Retail revenue / AWS profit	AWS 21.0% of sales	AWS and advertising	AI CapEx versus FCF
MSFT	Cloud and productivity	Balanced	Azure and Copilot	AI infrastructure return
GOOGL	Search	52.8%	Google Cloud	AI disruption to Search economics

This business comparison is essential to the drawdown framework. NVIDIA's historical volatility should be interpreted in the context of concentrated AI infrastructure exposure. Microsoft's stability reflects diversified recurring enterprise revenue. TSMC's threshold must be overlaid with a manufacturing cycle. Tesla's drawdown cannot be evaluated without separating current automotive economics from long-duration optionality.

3. Return and Drawdown Analysis

3.1 Long-Term Return and Risk Summary

Company	CAGR	Positive Years	Median Max Drawdown	Annualized Volatility	Downside Volatility	Typical Worst 1M	Typical Worst 3M	Median Recovery
NVDA	46.2%	68.8%	-28.7%	46.0%	27.9%	-17.7%	-15.1%	89.5 days
TSLA	44.6%	86.7%	-35.7%	54.1%	36.5%	-27.0%	-24.4%	204 days
AVGO	41.9%	93.8%	-22.9%	34.7%	24.6%	-19.0%	-14.6%	89.5 days
META	28.1%	84.6%	-19.7%	36.5%	24.3%	-15.3%	-14.0%	78 days
AAPL	26.5%	81.3%	-18.8%	26.5%	19.0%	-16.8%	-14.5%	100.5 days
TSM	26.4%	87.5%	-20.3%	26.8%	19.0%	-16.2%	-16.8%	130.5 days
AMZN	24.7%	81.3%	-21.2%	32.4%	21.2%	-17.6%	-13.7%	107 days
MSFT	21.0%	81.3%	-16.0%	23.1%	16.7%	-10.6%	-11.4%	80.5 days
GOOGL	20.7%	75.0%	-19.1%	28.0%	18.4%	-14.1%	-14.3%	121.5 days

3.2 How to Read the Columns Together

CAGR describes wealth creation, while Positive Years describes calendar-year consistency. Neither captures the losses experienced during the year. Median Max Drawdown adds the typical intra-year stress. Annualized Volatility describes the overall dispersion of returns, while Downside Volatility focuses on the negative portion. Worst 1M and 3M describe short-horizon pain. Median Recovery describes how long capital historically remained below a previous high.

No single column identifies the “best” stock. A high CAGR with high downside volatility can still be attractive, but it requires a smaller starting position and more dry powder. A lower CAGR with a shallow drawdown and fast recovery can support a larger permanent allocation and more continuous DCA.

3.3 CAGR Ranking: Wealth Creation

Rank	Company	CAGR
1	NVIDIA	46.2%
2	Tesla	44.6%
3	Broadcom	41.9%
4	Meta	28.1%
5	Apple	26.5%
6	TSMC	26.4%
7	Amazon	24.7%
8	Microsoft	21.0%
9	Alphabet	20.7%

NVIDIA, Tesla, and Broadcom dominate historical wealth creation. The ranking cannot be used mechanically for future allocation: the starting valuations, business mix, and scale of these companies are now very different from 2010. It does, however, show that accepting higher volatility was historically compensated in NVIDIA and Tesla, while Broadcom combined comparable compounding with a much smoother path.

3.4 Downside Volatility Ranking: Holding Difficulty

Rank, lowest risk first	Company	Downside volatility
1	Microsoft	16.7%
2	Alphabet	18.4%
3	Apple	19.0%
3	TSMC	19.0%
5	Amazon	21.2%

Rank, lowest risk first	Company	Downside volatility
6	Meta	24.3%
7	Broadcom	24.6%
8	NVIDIA	27.9%
9	Tesla	36.5%

Microsoft is the clearest low-downside-volatility compounder. Alphabet, Apple, and TSMC form the next tier. Tesla's downside volatility is more than twice Microsoft's. Consequently, equal dollar positions in Microsoft and Tesla do not represent equal behavioral or capital risk.

3.5 Recovery Ranking: Opportunity Cost of Being Underwater

Rank, fastest first	Company	Median recovery
1	Meta	78 days
2	Microsoft	80.5 days
3	NVIDIA	89.5 days
3	Broadcom	89.5 days
5	Apple	100.5 days
6	Amazon	107 days
7	Alphabet	121.5 days
8	TSMC	130.5 days
9	Tesla	204 days

Meta's median recovery was fastest despite occasional extreme tail events. NVIDIA's median recovery was also relatively fast, showing that its frequent corrections often occurred inside a powerful long-term trend. Tesla combined deep drawdowns with the longest recovery, increasing both psychological and opportunity cost. TSMC's longer recovery is consistent with a cyclical manufacturing business whose earnings and utilization cycles can take time to turn.

3.6 Typical Drawdown and Short-Horizon Loss Comparison

Microsoft's median annual maximum drawdown was only 16.0%, compared with 28.7% for NVIDIA and 35.7% for Tesla. Microsoft's typical worst one-month result was -10.6%; Tesla's was -27.0%. This explains why a drawdown-based strategy must be calibrated by stock. A 15% Microsoft decline already approaches a typical annual maximum; a 15% Tesla decline is ordinary noise.

Broadcom's combination is especially notable: 41.9% CAGR, 93.8% positive years, a 22.9% median annual drawdown, and an 89.5-day median recovery. Meta's normal-year metrics appear moderate, but its 30%–50% frequency later reveals heavier tail risk. TSMC's volatility is surprisingly close to Apple's, but its longer recovery and cyclical operating model require a different decision process.

3.7 Return Efficiency Heuristic

Rank	Company	CAGR / absolute median annual max drawdown
1	Broadcom	1.83
2	NVIDIA	1.61
3	Meta	1.42
4	Apple	1.41
5	Microsoft	1.31
6	TSMC	1.30
7	Tesla	1.25
8	Amazon	1.17
9	Alphabet	1.08

This is a non-standard descriptive heuristic, not a substitute for Sharpe, Sortino, valuation, or forward earnings analysis. Broadcom ranks first because it combined very high historical CAGR with a much smaller typical drawdown than NVIDIA or Tesla.

4. Drawdown Frequency and Historical Rarity

4.1 Historical Drawdown Recurrence

Company	at least 10%	at least 20%	at least 30%	at least 40%	at least 50%
NVDA	1.07 years	1.45 years	2.0 years	4.0 years	4.0 years
AAPL	1.07 years	2.29 years	4.0 years	Not observed	Not observed
GOOGL	1.23 years	2.29 years	5.33 years	16 years	Not observed
MSFT	1.23 years	4.0 years	16 years	Not observed	Not observed
AMZN	1.0 year	2.0 years	5.33 years	16 years	16 years
TSM	1.0 year	2.0 years	8.0 years	16 years	16 years
AVGO	1.07 years	1.45 years	4.0 years	8.0 years	Not observed
META	1.08 years	2.17 years	3.25 years	6.5 years	13 years
TSLA	1.0 year	1.0 year	1.25 years	3.0 years	7.5 years

Where the values come from

Each calendar year receives one annual maximum-drawdown observation. If a stock suffers several corrections during a year, only the deepest peak-to-trough decline determines which thresholds the year satisfies.

NVIDIA has 16 complete calendar years in the sample. Eight of those years contained a drawdown of at least 30%. Therefore, $16 / 8 = 2.0$ years. This means one out of every two observed calendar years qualified; it does not mean NVIDIA mechanically falls 30% once every two years.

Tesla has 15 complete years and all 15 contained a drawdown of at least 20%. Therefore, $15 / 15 = 1.0$ year. Microsoft had four qualifying 20% years out of 16, giving $16 / 4 = 4.0$ years.

Practical interpretation

Recurrence interval	Historical interpretation
Around 1 year	Close to an annual norm
1–2 years	Frequent
2–3 years	Meaningful but recurring correction
3–6 years	Large drawdown
6–10 years	Rare drawdown
More than 10 years	Historical tail event
Not observed	Outside this annual sample, not impossible

At the 20% threshold, Tesla is the extreme: the decline was universal in the sample. NVIDIA and Broadcom are also frequent at 1.45 years. Amazon and TSMC are approximately once every two years. Microsoft is the outlier at once every four years.

At 30%, the separation becomes more useful. Tesla remained very frequent at 1.25 years; NVIDIA was once every two years. Meta was once every 3.25 years. Apple and Broadcom were once every four years. Alphabet and Amazon were once every 5.33 years. TSMC was once every eight years and Microsoft once in 16 years.

At 40%, a drawdown was still observed every three years for Tesla and every four years for NVIDIA. Broadcom's interval lengthened to eight years. Alphabet, Amazon, and TSMC each recorded one qualifying annual observation in 16 years. Apple and Microsoft did not record one.

4.2 Annual Historical Probability of Major Drawdowns

Company	at least 20%	at least 30%	at least 40%
TSLA	100.0%	80.0%	33.3%
NVDA	68.8%	50.0%	25.0%
AVGO	68.8%	25.0%	12.5%
AMZN	50.0%	18.8%	6.3%
TSM	50.0%	12.5%	6.3%

Company	at least 20%	at least 30%	at least 40%
META	46.1%	30.8%	15.4%
AAPL	43.7%	25.0%	0.0%
GOOGL	43.7%	18.8%	6.3%
MSFT	25.0%	6.3%	0.0%

Probability and recurrence are the same historical counts expressed differently. Eight qualifying years out of 16 produce both a 50% historical frequency and a two-year recurrence interval.

Company comparison at 20%

- **Tesla: 100%.** A 20% decline was ordinary, not an exceptional entry signal.
- **NVIDIA and Broadcom: 68.8%.** Both frequently corrected 20%, but their deeper tails differed.
- **Amazon and TSMC: 50%.** A 20% decline was meaningful but still occurred in half the years.
- **Meta, Apple, and Alphabet: approximately 44%–46%.** A 20% correction justifies faster accumulation, not necessarily maximum deployment.
- **Microsoft: 25%.** A 20% decline carried the greatest rarity and therefore the strongest relative price signal in this group.

Company comparison at 30%

Tesla remained common at 80%, and NVIDIA at 50%. Meta's 30.8% demonstrates its heavier tail relative to Apple and Alphabet. Broadcom and Apple were both 25%. Amazon and Alphabet were 18.8%. TSMC fell sharply to 12.5%, showing that many TSMC corrections stopped between 20% and 30%. Microsoft's 6.3% makes a 30% decline a historical tail event.

Broadcom versus NVIDIA

Both stocks had a 68.8% frequency at 20%. At 30%, NVIDIA remained at 50% while Broadcom fell to 25%. At 40%, NVIDIA was 25% and Broadcom 12.5%. Broadcom therefore experienced frequent corrections but fewer severe extensions, consistent with its superior return-to-typical-drawdown heuristic.

Apple and Alphabet versus Meta

Their 20% frequencies were similar. The tails were not. Meta reached 30% in 30.8% of years and 40% in 15.4%. Apple reached 30% in 25% and did not record 40%; Alphabet reached 30% in 18.8% and 40% in 6.3%. Meta therefore requires more capital reserved for tail events and more rigorous thesis checks when they occur.

Statistical limitation

These values are historical sample frequencies, not forward-looking probabilities. "Tesla 100%" does not guarantee another 20% decline next year. "Microsoft 0% at 40%" does not make a future 40% decline impossible. The sample contains only 13–16 complete years per company and is most useful for relative comparison.

5. Integrated Stock-by-Stock Investment Strategy

5.1 Microsoft — Core DCA Compounder

Microsoft combines the group's most balanced business mix with the lowest downside volatility and shallowest median annual drawdown. Intelligent Cloud grew 32%, Azure grew 43%, and Productivity and Business Processes grew 14%, while the smaller consumer segment declined. This diversification supports recurring cash flow but does not eliminate the emerging risk that AI infrastructure spending outruns monetization.

Historically, Microsoft produced a 21.0% CAGR, 16.0% median annual drawdown, 16.7% downside volatility, and an 80.5-day median recovery. A 20% drawdown occurred in only 25% of years, while 30% occurred once in the 16-year sample.

Strategy: maintain full continuous DCA. Treat approximately 15% as normal-to-moderate volatility, 20% as the first major acceleration zone, and 30% as a rare major accumulation zone. At 20%, consider approximately 2x the normal

contribution. At 30%, approximately 3x can be justified if Azure demand, Microsoft 365 retention, Copilot monetization, cloud margins, and free-cash-flow returns on AI CapEx remain intact.

5.2 Alphabet — Core DCA with Search-Thesis Validation

Alphabet's Search business still contributes 52.8% of revenue and grew 16.8%, while Cloud contributes 20.7% and grew 81.8%. This combination creates a powerful second engine, but the principal risk is whether generative AI changes Search monetization or cost-to-serve.

Historically, Alphabet produced a 20.7% CAGR, 19.1% median annual drawdown, 18.4% downside volatility, and a 121.5-day median recovery. A 20% drawdown occurred in 43.7% of years; 30% in 18.8%; and 40% in 6.3%.

Strategy: maintain continuous DCA. Increase to approximately 1.5x near 20% and 2.5x near 30%. A 40% decline may justify approximately 3x only after verifying query growth, AI Search monetization, Google Cloud growth and margin, YouTube, CapEx, and free cash flow. A deep decline specifically caused by structural Search impairment is not automatically an opportunity.

5.3 Apple — Core Quality Compounder

Apple's iPhone still contributes 49.6% of revenue, while Services contributes 28.1% and provides the most important recurring, high-margin layer. Mac and iPhone led the latest growth; iPad declined. The thesis rests on the installed base, retention, Services economics, product replacement, China, and AI competitiveness.

Historically, Apple produced a 26.5% CAGR, 18.8% median annual drawdown, 19.0% downside volatility, and a 100.5-day median recovery. A 20% decline occurred in 43.7% of years and 30% in 25%; 40% was not observed.

Strategy: continuous DCA. Increase to approximately 1.5x around 20% and 2.5x around 30% if installed-base engagement, Services growth, product gross margins, and cash generation remain intact. A decline approaching 40% lies outside this annual sample and should trigger a full thesis and valuation review rather than automatic buying.

5.4 Broadcom — High-Return Opportunistic Compounder

Broadcom's semiconductor business contributes 67.6% of revenue and grew 79%. AI semiconductor revenue grew 143% and now approaches half of total revenue. Infrastructure Software provides slower 9% growth but recurring cash flow. The combination explains Broadcom's exceptional historical return efficiency, but exposure to a few hyperscalers and AI programs increases cycle risk.

Historically, Broadcom produced a 41.9% CAGR, 93.8% positive years, a 22.9% median annual drawdown, and an 89.5-day recovery. A 20% decline was frequent at 68.8%, but 30% fell to 25% and 40% to 12.5%.

Strategy: maintain a 0.75x–1x baseline contribution and reserve 20%–30% of planned capital as dry powder. A 20% correction deserves only a modest increase to roughly 1.25x. Approximately 30% is the first major add zone at roughly 2x. Approximately 40% can justify 3x staged deployment after validating AI accelerator wins, networking demand, customer concentration, VMware retention, leverage, and free cash flow.

5.5 Meta — Advertising Compounder with Tail-Risk Discipline

Meta receives 97.6% of revenue from advertising. AI is currently most valuable when it improves content recommendation, engagement, advertiser conversion, and pricing. Reality Labs remains economically small and deeply loss-making. Meta's fundamental risk is therefore not revenue diversification but whether the advertising engine can fund rising AI infrastructure and long-duration experimental investments.

Historically, Meta produced a 28.1% CAGR, 19.7% median annual drawdown, and the fastest 78-day median recovery. However, 30% and 40% drawdowns occurred in 30.8% and 15.4% of years, revealing heavier tail risk than the median suggests.

Strategy: maintain a 0.75x–1x baseline. Increase to approximately 1.5x near 20% and 2x near 30%. At 40%, 2.5x–3x is appropriate only after mandatory review of engagement, ad impressions, ad pricing, advertiser ROI, Reality Labs losses, CapEx, operating margin, and free cash flow. At 50% or deeper, suspend mechanical averaging and rebuild the thesis.

5.6 Amazon — AWS-Led Opportunistic Compounder

Retail generates most revenue, but AWS generates approximately 60% of operating income and grew 37%. Advertising grew 26%. Amazon's value depends increasingly on whether cloud and advertising profitability can offset the capital intensity of logistics and AI infrastructure.

Historically, Amazon produced a 24.7% CAGR, 21.2% median annual drawdown, 21.2% downside volatility, and a 107-day median recovery. A 20% drawdown occurred in half of years, 30% in 18.8%, and 40% in 6.3%.

Strategy: maintain 0.75x–1x continuous exposure. Increase to approximately 1.5x near 20%, 2.5x near 30%, and potentially 3x near 40%. Before major deployment, verify AWS growth, AWS margins, AI revenue, advertising growth, retail margin, international profitability, CapEx, and free-cash-flow conversion.

5.7 TSMC — Cyclical Compounder

HPC now represents 66% of revenue and smartphones only 22%. Advanced processes generate 77% of wafer revenue. This confirms that AI and leading-edge computing are the principal growth drivers, but TSMC remains a capital-intensive manufacturer exposed to utilization, customer inventories, pricing, and geopolitical risk.

Historically, TSMC produced a 26.4% CAGR with Apple-like downside volatility of 19.0%, but its median recovery was longer at 130.5 days. A 20% decline occurred in 50% of years, while 30% occurred in only 12.5%.

Strategy: use a lighter 0.5x–0.75x base DCA. Approximately 20% is an initial add zone at 1.5x, but it remains normal for the stock. Approximately 30% is historically much more significant and can justify 2.5x–3x after reviewing HPC demand, node utilization, N2 execution, advanced packaging, gross margins, CapEx, customer concentration, and geopolitics. At 40%, conduct a full cycle and geopolitical review.

5.8 NVIDIA — High-Beta AI Compounder

NVIDIA receives 92.1% of revenue from Data Center, with extraordinary growth in both compute and networking. The same concentration that produces exceptional growth also makes the stock sensitive to AI infrastructure expectations, customer CapEx, utilization, competition, and gross margin.

Historically, NVIDIA produced the highest CAGR at 46.2%, but its median annual drawdown was 28.7% and downside volatility was 27.9%. A 20% decline occurred in 68.8% of years, 30% in 50%, and 40% in 25%.

Strategy: maintain a smaller 0.5x–0.75x base contribution. Do not deploy major dry powder at 20%; it is normal. Approximately 30% is the first meaningful accumulation zone at 1.5x–2x. Approximately 40% is the major zone at 2.5x–3x if hyperscaler CapEx, GPU demand and utilization, CUDA, networking, HBM supply, competitive position, and gross margins remain intact. At 50%, conduct a full cycle review before continuing.

5.9 Tesla — High-Beta Optionality Asset

Approximately 72.7% of Tesla's latest revenue remains automotive. Energy contributes 11.1% and Services 16.2%. Autonomy, Robotaxi, and Optimus may carry substantial valuation, but investors should distinguish that optionality from demonstrated revenue and cash flow.

Historically, Tesla produced a 44.6% CAGR but also a 35.7% median annual drawdown, 36.5% downside volatility, a -27.0% typical worst one-month result, and a 204-day median recovery. Every observed year contained a 20% drawdown, 80% contained 30%, and one third contained 40%.

Strategy: maintain only a 0x–0.5x base contribution. Do not add incrementally merely because the stock falls 20%. A 30% decline still supports only normal or modest deployment. Approximately 40% is the first meaningful zone at 1.5x–2x. Approximately 50% may justify 2.5x after reviewing deliveries, automotive gross margin, pricing, FSD adoption and economics, Robotaxi regulation, Energy, free cash flow, CapEx, and balance-sheet strength. At 60% or deeper, rebuild the valuation and thesis from the ground up.

6. Portfolio Classification and Entry Framework

6.1 Which Companies Are Best Suited to Continuous Long-Term Ownership?

Highest baseline DCA intensity

Microsoft, Alphabet, and Apple are the strongest continuous-DCA candidates. They combine diversified or entrenched business models with comparatively moderate downside volatility. Waiting only for a 30%–40% crash risks long periods of underinvestment.

- Microsoft: continuous 1x DCA; accelerate at 20%; major add at 30%.
- Alphabet: continuous 1x DCA; accelerate at 20%; major add at 30%, with Search review.
- Apple: continuous 1x DCA; accelerate at 20%; major add at 30%.

Continuous ownership with more reserved capital

Broadcom, Meta, and Amazon merit baseline positions, but their higher tail or cycle risks create more value in holding dry powder.

- Broadcom: 0.75x–1x base; first major add at 30%; strongest opportunity near 40% if AI demand remains intact.
- Meta: 0.75x–1x base; add at 20% and 30%; mandatory thesis review at 40%.
- Amazon: 0.75x–1x base; add at 20%; major add at 30%; strongest opportunity near 40% if AWS and FCF remain healthy.

6.2 Which Companies Are Better Bought Opportunistically?

TSMC, NVIDIA, and Tesla require lower baseline intensity because ordinary volatility is larger or the business has a stronger cycle/optionality component.

- TSMC: light base ownership; 20% is an initial add; 30% is the major cycle-adjusted opportunity.
- NVIDIA: light base ownership; do not treat 20% as exceptional; first meaningful add at 30%; major add at 40%.
- Tesla: minimal base; 20%–30% is normal; first meaningful add at 40%; major thesis-dependent add at 50%.

6.3 Suggested Drawdown Allocation Matrix

Assume 1x is the normal planned contribution to a stock, not a target portfolio weight.

Company	Base DCA	-20%	-30%	-40%	-50%	Classification
MSFT	1x	2x	3x	Thesis review	—	Core DCA
GOOGL	1x	1.5x	2.5x	3x*	—	Core DCA
AAPL	1x	1.5x	2.5x	Thesis review	—	Core DCA
AVGO	0.75x–1x	1.25x	2x	3x*	—	Opportunistic compounder
META	0.75x–1x	1.5x	2x	2.5x–3x*	Thesis review	Opportunistic compounder
AMZN	0.75x–1x	1.5x	2.5x	3x*	Thesis review	Opportunistic compounder
TSM	0.5x–0.75x	1.5x	2.5x–3x*	Thesis review	Thesis review	Cyclical compounder
NVDA	0.5x–0.75x	1x	1.5x–2x	2.5x–3x*	Full cycle review	High-beta compounder
TSLA	0x–0.5x	No incremental add	1x	1.5x–2x*	2.5x*	Optionality asset

* Requires confirmation that the long-term business thesis and valuation remain attractive.

6.4 Portfolio-Level Dry Powder

Dry powder should be assigned according to the probability that a stock will present deeper entry points.

Lower dry-powder requirement: Microsoft, Alphabet, Apple. Deep drawdowns are less frequent, so excess cash can create significant opportunity cost.

Moderate dry-powder requirement: Broadcom, Meta, Amazon, TSMC. Maintain ownership but reserve capital for 30%–40% corrections.

Higher dry-powder requirement: NVIDIA, Tesla. Because 20%–30% declines are common, deploying all capital early can leave the investor unable to act when the drawdown becomes genuinely unusual.

6.5 Final Decision Rule

A drawdown qualifies as a strong accumulation opportunity only when four tests align:

1. **Business quality:** the company retains a durable competitive advantage.
2. **Thesis integrity:** the reason for owning the company remains valid.
3. **Drawdown rarity:** the decline is unusually deep relative to that stock's history.
4. **Valuation:** the new price materially improves forward expected return.

A rare drawdown caused by temporary macro fear can be an opportunity. A rare drawdown caused by permanent impairment can be a value trap.

7. Limitations and Future Extension

The historical sample is short for estimating true tail probabilities. Annual maximum drawdown also records only one observation per calendar year and does not count independent corrections. A future event-based extension should identify each distinct peak-to-trough episode and measure event frequency, duration, recovery, clustering, and macro regime.

The business-mix tables use different fiscal-quarter endpoints. Segment definitions also differ: revenue shares are comparable as concentration indicators, not as identical accounting categories. TSMC reports platform growth sequentially, while most other tables use year-over-year growth. Tesla's automotive figure is derived from total revenue less the two separately disclosed non-automotive categories and is therefore shown as approximate.

Finally, historical CAGR should not be treated as a forward-return forecast. Several companies are now much larger and trade at different valuations than they did at the beginning of the sample.

8. Conclusion

The nine companies should not be managed with one universal dip-buying rule.

Microsoft, Alphabet, and Apple are best treated as permanent core compounders. Their business durability and moderate historical downside favor continuous DCA, with faster deployment around 20% and major accumulation around 30%.

Broadcom, Meta, and Amazon are also long-term compounders, but their cycle, concentration, or tail risks justify retaining more capital for 30%–40% corrections.

TSMC is a cyclical compounder. Its historical price behavior is relatively stable, but drawdown decisions must be connected to advanced-node demand, utilization, capital expenditure, and geopolitics.

NVIDIA is a high-beta AI compounder. A 20% decline is ordinary; 30% is the first meaningful threshold and 40% is the major accumulation zone if AI demand and competitive advantages remain intact.

Tesla is an optionality-heavy asset. A 20%–30% decline is not statistically unusual. Meaningful incremental buying begins closer to 40%, while 50% requires detailed validation of both current automotive economics and future autonomy, robotics, and energy value.

The objective is not to predict the exact bottom. It is to convert volatility into a disciplined capital-allocation process: understand what the company earns money from, measure how unusual the drawdown really is, confirm that the business thesis survives, and increase investment intensity only when price and fundamentals align.

8.1 Primary Financial Sources

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- Microsoft, FY2026 Q4 results, July 2026.
- Alphabet, Q2 2026 earnings release, July 22, 2026.